The Evolution of Data: From Databases to Databricks

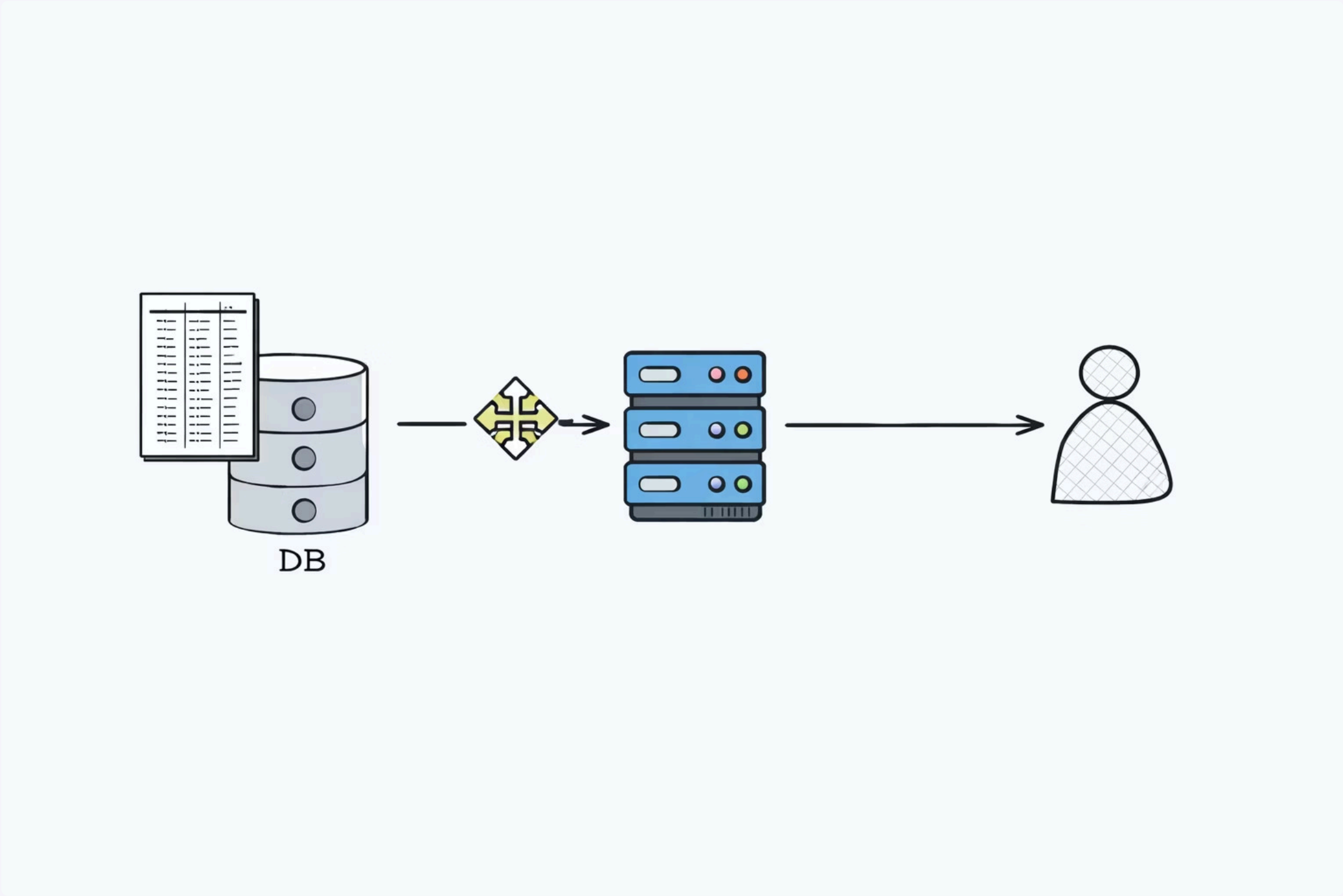
Presentation Content for Gamma App

The Beginning — Single Machine Databases

In the early days of data, everything ran on a single machine. The architecture was straightforward:

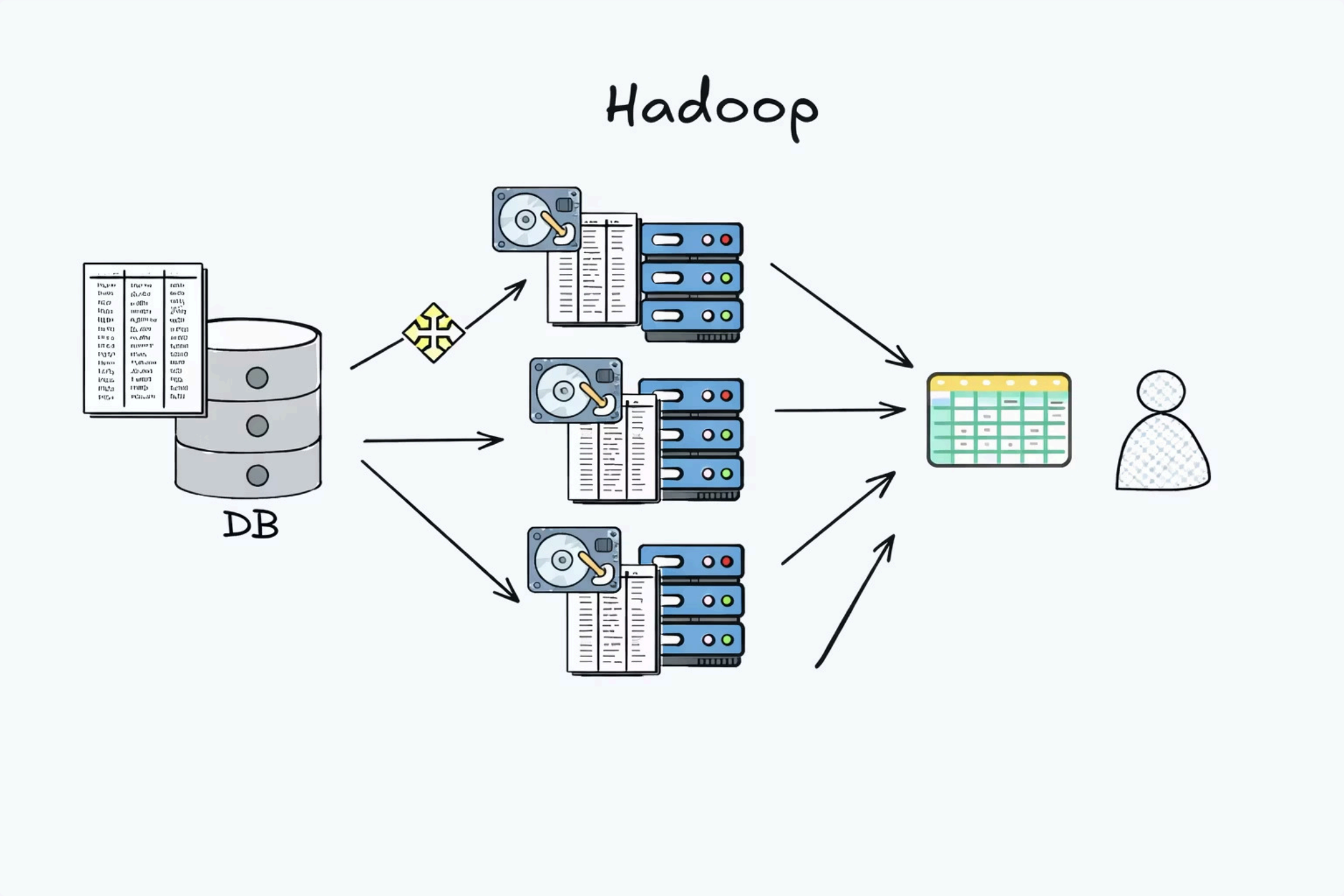
Database → ETL Process → Application Server → End User

A traditional relational database (Oracle, SQL Server, MySQL) stored all the data. An ETL (Extract, Transform, Load) process pulled data, transformed it, and loaded it into a single server that served the end user.



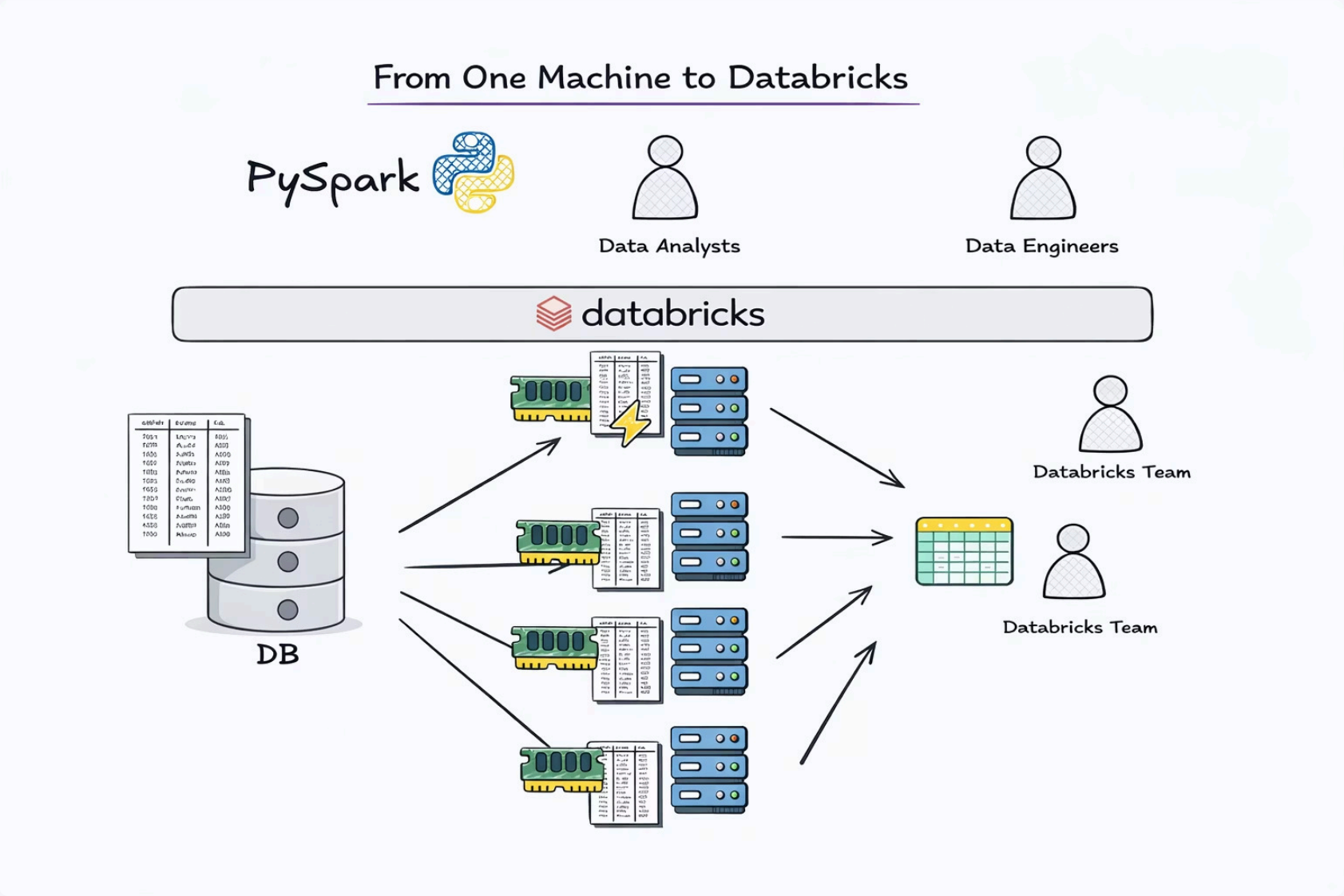
Enter Hadoop — Distributed Processing on Disk

To solve the performance bottleneck, **Hadoop** introduced the concept of distributed computing. Instead of one machine, data was split across multiple nodes.



From One Machine to Databricks — The Spark Revolution

Apache Spark and **PySpark** changed everything by introducing **in-memory processing**.



The Spark Advantage:

- Data is loaded into **RAM** and processed in-memory, up to **100x faster** than Hadoop MapReduce
- Supports Python (PySpark), SQL, Scala, R, accessible to both engineers and analysts
- Unified engine for batch, streaming, SQL, and ML

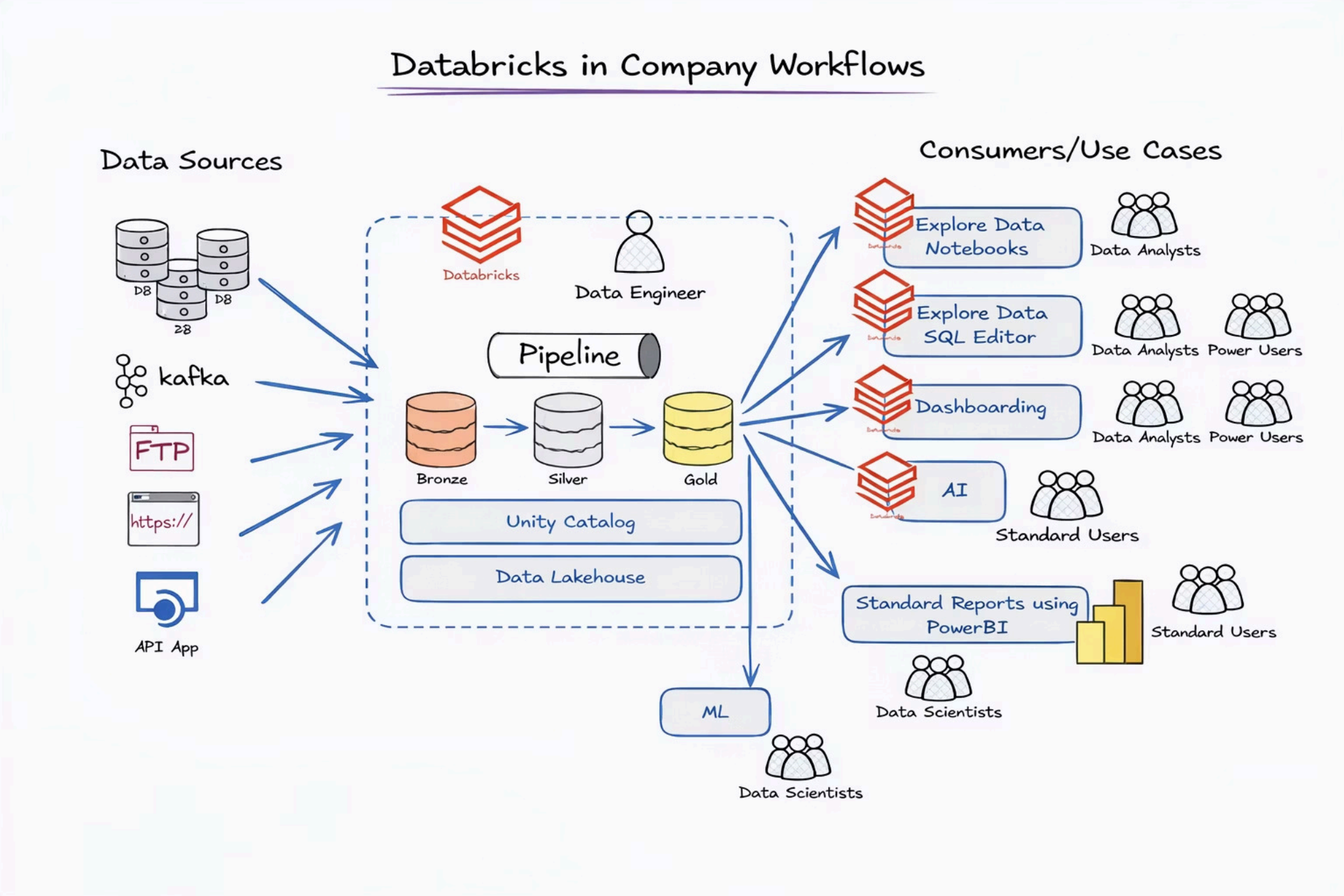
But the industry still had a problem:

- Data Analysts and Data Engineers worked in **separate silos**
- Different tools, different environments, different languages
- A lot of staff required, data engineers to build pipelines, analysts to query, scientists to model
- No unified governance or catalog across teams

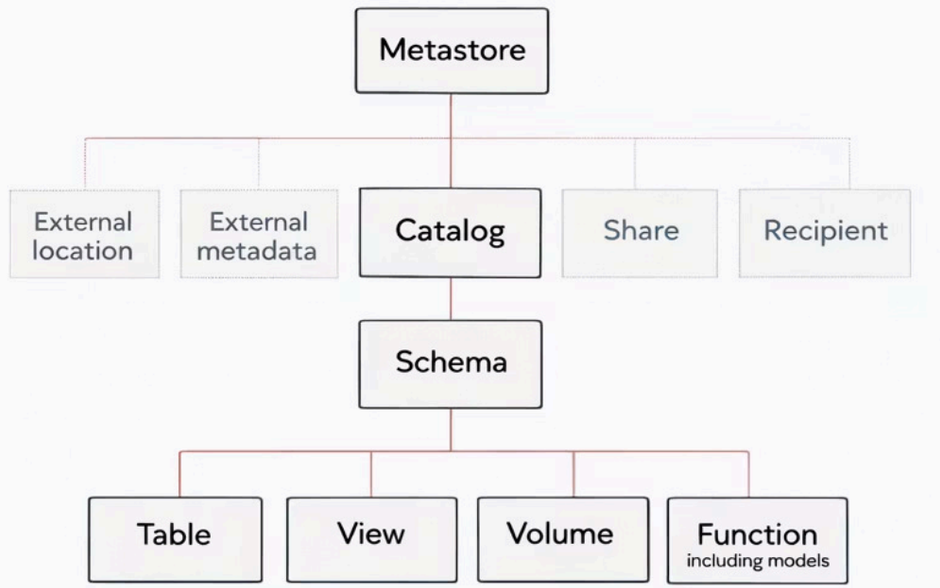
The Unified Platform — Azure Databricks Lakehouse

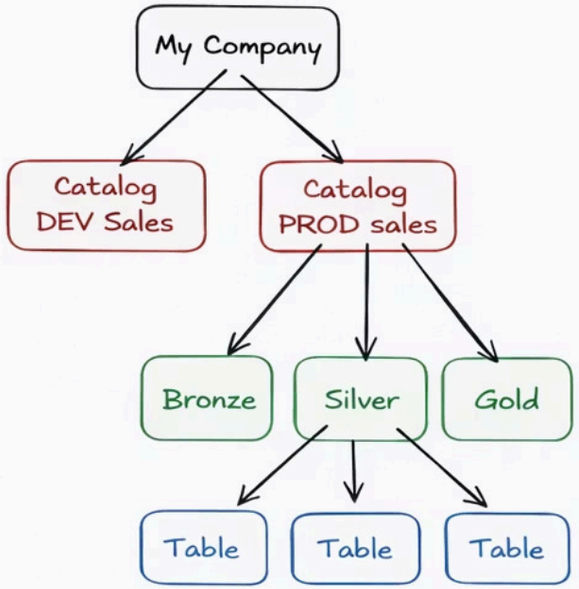
Databricks introduced the **Lakehouse** architecture, combining the best of data lakes and data warehouses.

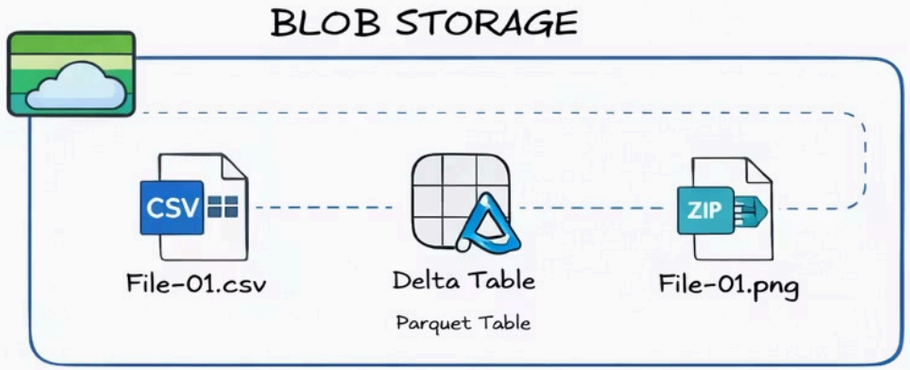
End-to-End Data Platform in Action



Unity Catalog & Blob Storage

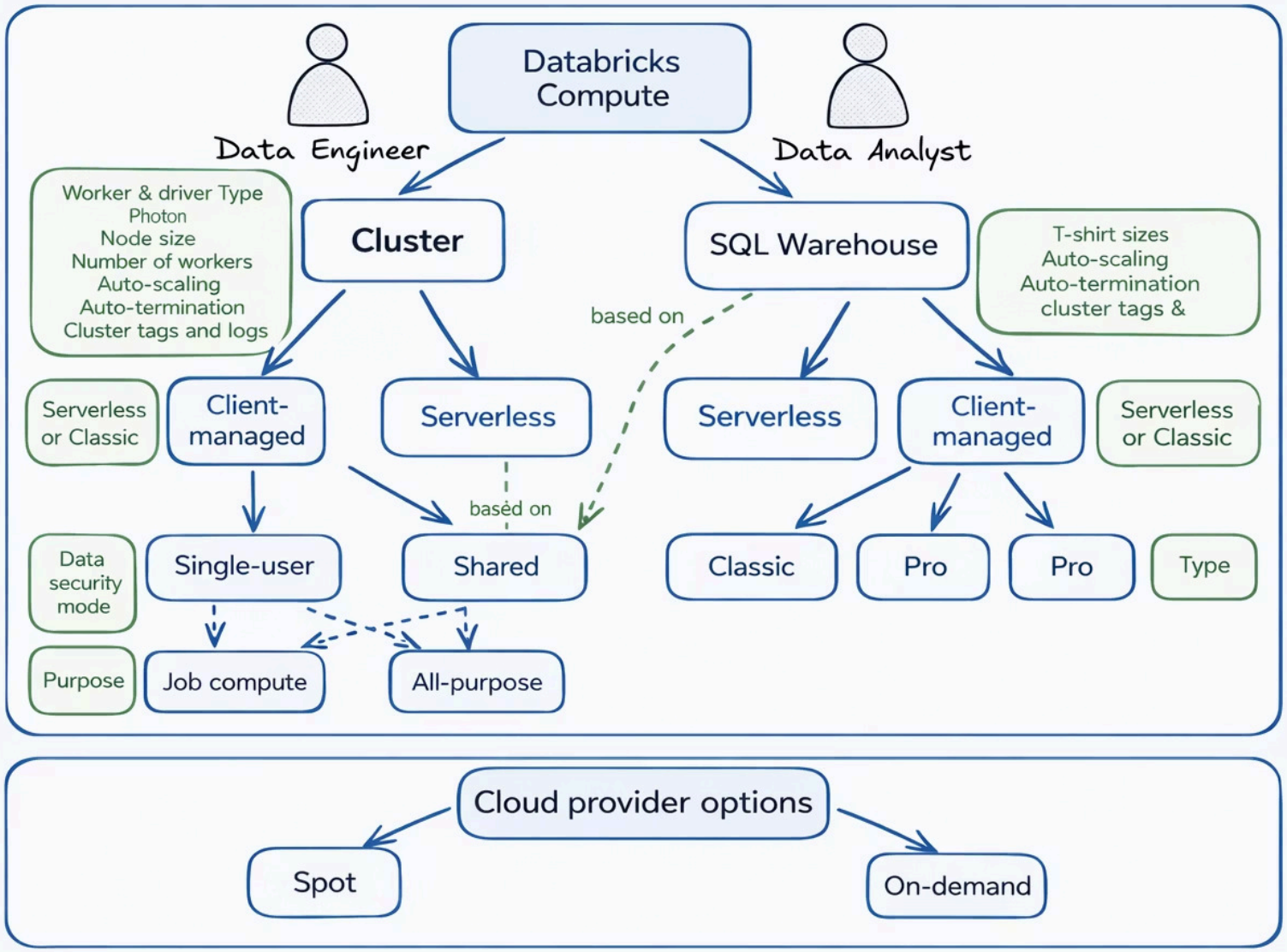




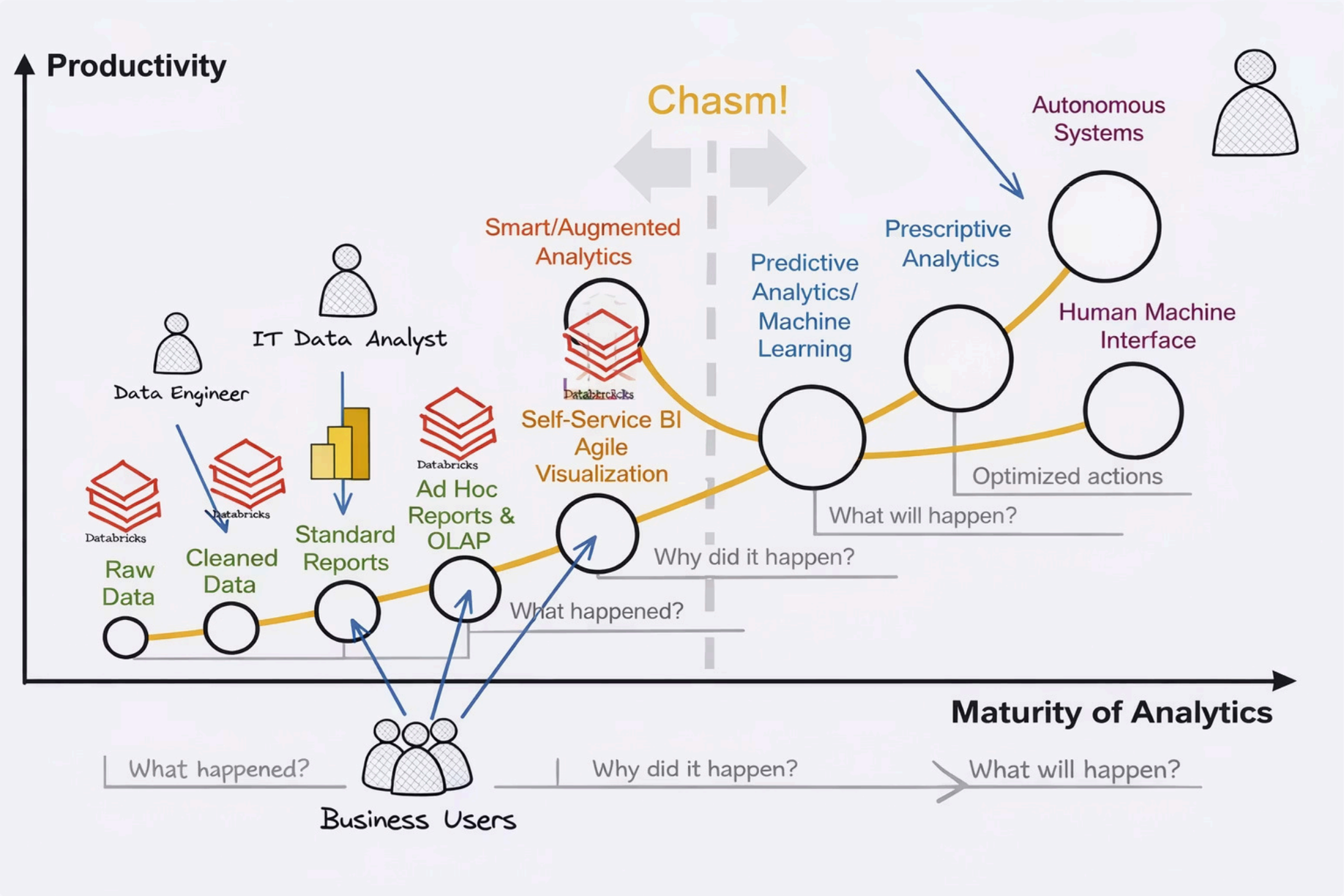


Understanding Databricks Compute — Clusters & SQL Warehouses

Databricks offers two main compute options depending on your role and workload.

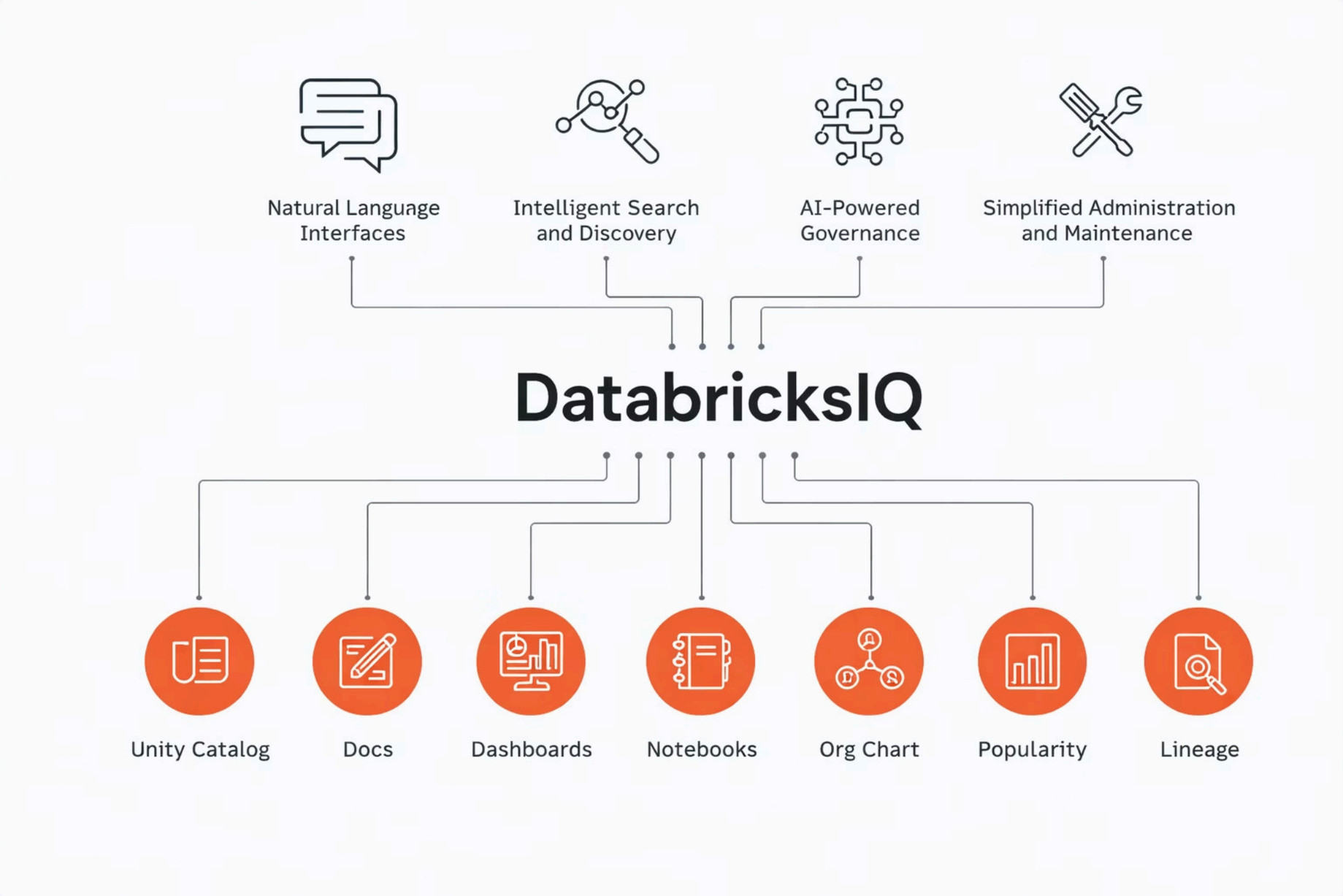
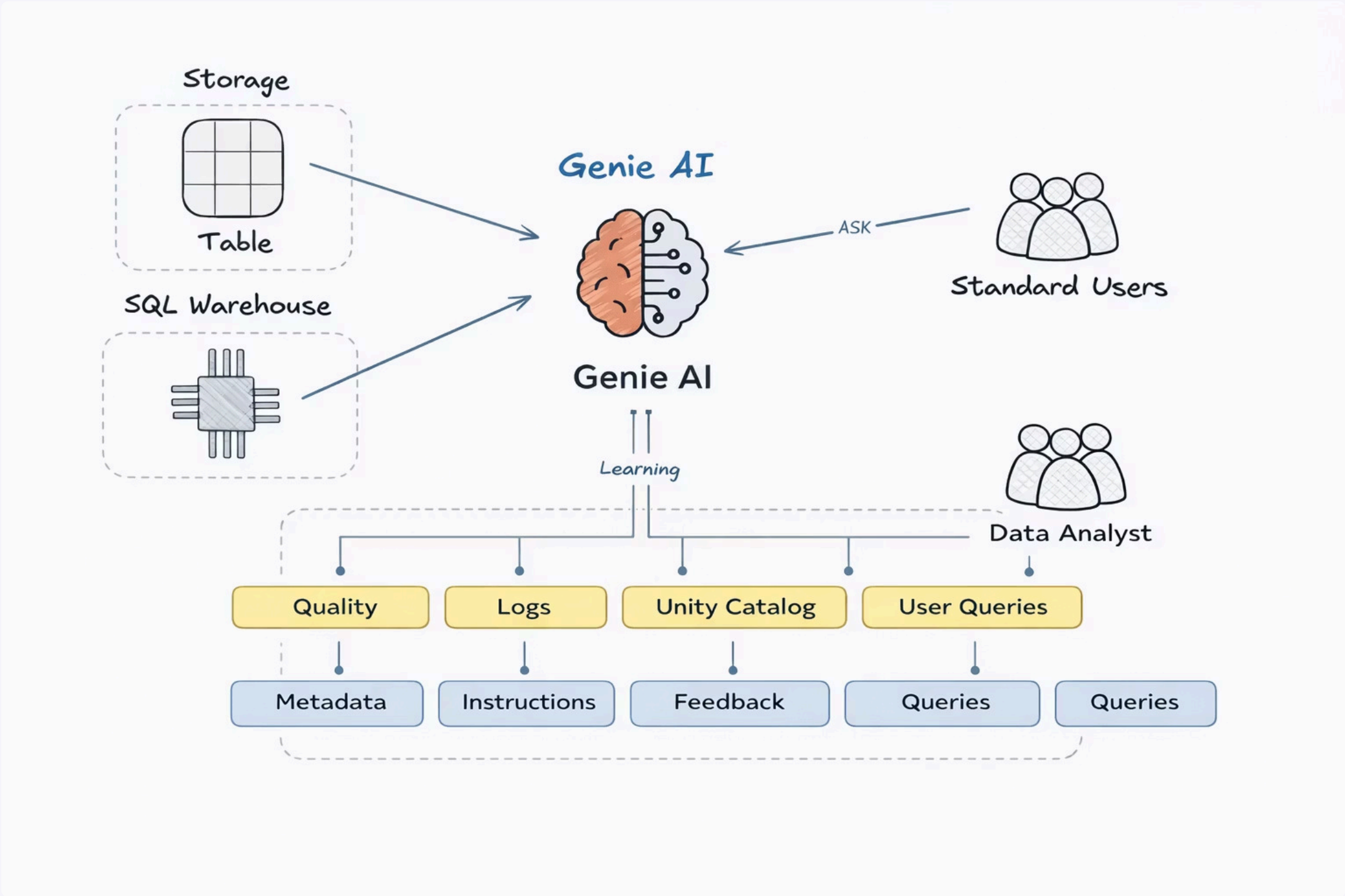


Analytics Maturity Journey — Where Does Your Organization Sit?



DatabricksIQ — The AI Brain of the Lakehouse

DatabricksIQ is Databricks' AI layer that understands your data, your organization, and your usage patterns.



The Journey — From Databases to Databricks